

BluHorizon AI - Full-Stack RAG Application

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Overview

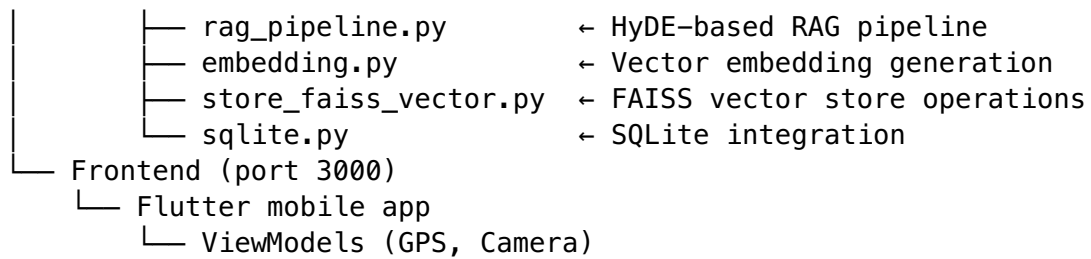
BluHorizon AI is a full-stack Retrieval-Augmented Generation (RAG) system with a Flutter mobile frontend and a Python FastAPI backend. Users can ingest documents (PDF, DOCX, XLSX), which are embedded into a FAISS vector store, then query them through an AI-powered chat interface backed by Claude.

Technologies

Layer	Stack
Frontend	Flutter (Dart), mobile-first
Backend	Python, FastAPI, Uvicorn
AI	Anthropic Claude API
Vector DB	FAISS (CPU)
Embeddings	Sentence-Transformers
Database	Firebase, SQLite
ML	PyTorch, tiktoken
Search	Tavily API
Deployment	Docker, Docker Compose

Architecture

```
docker-compose.yml
├─ Backend (port 8000)
│   └─ routers/           ← chat, RAG, agents endpoints
│       └─ services/
│           └─ claude_client.py    ← Anthropic API wrapper (tool support)
```



Branch Context

The repository keeps the stable demo and the database migration work separated:

Branch	Purpose
main	Stable portfolio/demo branch for the current FAISS-based RAG application with SQLite/Firebase-era persistence pieces.
postgres-migration	Work-in-progress branch for the Postgres migration, JWT/auth flow, Flutter login screen, Docker updates, and Postgres-backed storage refactor.

This separation keeps the original demo reviewable while the production-style database/auth work is tested before merge.

Key Features

- **HyDE RAG Pipeline:** Generates a hypothetical answer to the user query, embeds it, and uses it for semantic retrieval — improving recall over naive keyword search
- **Gap-aware chunk filtering:** Dynamic score-gap cutoff selects the most relevant document chunks rather than using a fixed top-K
- **Document ingestion:** Supports PDF, DOCX, and XLSX with configurable chunking strategies
- **Tool-augmented Claude:** Custom tools (web search, code execution) extend Claude’s capabilities within the chat interface
- **Mobile-native features:** GPS and camera ViewModels integrated into the Flutter frontend
- **Containerized deployment:** Single docker-compose up launches both backend and frontend

Team Contributions

- **Diego:** ViewModels (GPS and Camera integration)
- **Payton:** UI/UX design
- **Jacky + Connor:** Firebase / database layer

RAG Pipeline Flow

User query

- HyDE: Claude generates hypothetical document
- Embed hypothetical doc with Sentence-Transformers
- FAISS similarity search
- Gap-aware chunk selection
- Retrieved chunks + original query → Claude response